

The First Foods Initiative

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Summary

For most of human history, the first solid foods that infants ate were fermented. Fermented porridges and gruels, from *ogi* in West Africa to *poi* in Polynesia, were near-universal weaning foods. These foods were a dense source of live microbes, microbial metabolites, and fermented fibers, three classes of bioactives that are independently known to shape gut microbial assembly, immune-cell signaling, and tolerance induction. Delivered during the first 1000 days of life, these food-based bioactives would have been ideally timed to shape the developing immune system. Within the past century, industrialization eliminated almost all of these foods from infant diets in high-income countries. Over the same period, and possibly not coincidentally, childhood atopic disease (food allergy, asthma, and eczema) has risen to affect more than 1 in 4 U.S. children.

The First Foods Initiative is a \$35M, five-year philanthropic fund built on a specific hypothesis: that the disappearance of fermented foods from early-life diets is a significant contributor to rising childhood atopic disease, and that restoring early-life fermented food exposure could meaningfully reduce that disease burden.

The Challenge

Childhood atopic disease has risen dramatically over the past three decades. In the US, childhood food allergy alone affects roughly 4 million children and costs an estimated \$24.8B per year, a figure that does not capture the lifelong burden these conditions place on the children who develop them. The dominant scientific framework for understanding this rise (the hygiene hypothesis and its descendants in microbiome science) holds that reduced microbial exposure during early-life immune development drives a loss of tolerance. But one major source of early-life microbial and bioactive exposure has remained underexplored: food itself, and fermented foods in particular.

Until the 20th century, fermentation was a dominant method of food preparation in nearly every traditional food culture, and fermented cereal and starch-based porridges were common weaning foods around the world. Over the past century, industrialization replaced these foods with sterilized, shelf-stable alternatives: pasteurized formula, refined infant cereals, and jarred purees. Observational evidence suggests fermented food consumption may protect against atopic disease (30–70% risk reductions with infant or maternal yogurt consumption), but the mechanistic, causal, and clinical trial data needed to act on that signal don't yet exist.

Both scientific and structural barriers have kept the field from generating and translating that data. As a result, foundational science across the food-immune intersection has been underfunded, and the multi-sector coordination needed to move findings into practice has had no clear home.

Why Now

The mechanistic tools are mature. Multi-omics platforms can now resolve fermented foods and the immune responses they drive, and machine learning makes it possible to integrate these layers to identify connections at scale. My team already applied these tools to over 100 fermented foods to characterize their impact on inflammatory pathways in immune cells, and we can expand on this dataset to identify drivers of immune development.

The model systems are aligned with the right questions. Identifying the fermented food components that impact immune development requires a tiered strategy: primary immune cell assays to triage candidates from large bioactivity screens, infant-derived cell-based and organoid systems for mechanistic follow-up, and mouse models for *in vivo* validation in the weaning window. Each layer has matured in the past two to three years. The fund's job is to coordinate them around a single thesis, rather than to develop any one from scratch.

The observational signal is consistent. Multiple birth-cohort studies now show consistent, dose-responsive associations between yogurt consumption in infancy or pregnancy and reduced atopic disease risk. A 2021 Stanford trial extended this from observation to intervention in adults, which showed significant reduction in markers of inflammation with increased fermented food consumption. The pattern is consistent enough that the question is no longer whether fermented foods are capable of affecting immune status, but how to translate that effect into a pediatric intervention.

The translational paradigm has been validated. The LEAP trial demonstrated that the right intervention in the right developmental window (early exposure to peanut allergens) can reshape global pediatric guidelines within two years. The pathway from RCT to AAP guidance to clinical practice is now well-mapped, and the same translational route is open to fermented foods once the science is built.

The Plan

The First Foods Initiative is a \$35M, five-year philanthropic fund that will build the scientific and translational foundation for fermented foods as an evidence-based strategy to prevent childhood atopic disease. The work will be structured in two coordinated phases.

Phase 1 (Years 1–3, \$20M): Discovery and causal validation

Phase 1 will establish the mechanistic foundation and narrow the search space for future clinical work. Funded work will include:

- **An integrated discovery pipeline** that builds on existing multi-omic characterization of 100+ globally important fermented foods (Microcosm Foods / PTFI), extending it with a tiered screening platform for early-life immune biology, causal perturbation studies to identify which microbes, metabolites, and fibers drive tolerance, and machine-learning integration across the layers.
- **The Fermented Foods Atlas**, the flagship open output of this pipeline, linking food composition to functional immune readouts.
- **Pilot translational studies** examining fermented food introduction during complementary feeding, in collaboration with established pediatric cohorts.

Phase 2 (Years 4–5, \$15M): Clinical translation and deployment

Phase 2 seeds the clinical and deployment effort that should follow if Phase 1 succeeds. Phase 2 will build the evidence, partnerships, and infrastructure needed to attract next-stage funding. Each of the activities below targets a specific handoff in the path from evidence to practice:

- **Pilot pediatric trials → evidence base for guidelines.** Trials in infants at elevated atopic risk, generating the safety, feasibility, and preliminary efficacy data needed to power and motivate larger Phase 3 RCTs.
- **Engagement with pediatric societies → guideline review.** Working with the AAP, WHO, and pediatric allergy societies to translate emerging evidence into dietary guidance, modeled on the post-LEAP guideline revisions for early peanut introduction.

- **Industry partnerships → accessible products.** Product translation pathways with food companies to develop safe, affordable, infant-appropriate fermented foods, designed for accessibility rather than premium positioning.
- **Policy and food-system engagement → reach.** Pediatric feeding curricula and regulatory pathways for live-microbe infant foods to ensure that evidence translates into practice for the families who need it most.

What success looks like. If the hypothesis holds, the Initiative will produce **1)** the first systematic, open datasets linking fermented foods to early-life immune biology, **2)** causal evidence identifying which specific components drive tolerance and which do not, **3)** pilot clinical evidence to motivate larger trials and inform pediatric dietary guidance, and **4)** open pathways for industry hand-offs with the supporting regulatory groundwork.

Falsification and go/no-go criteria. If Phase 1 finds no causal signal linking fermented food bioactives to immune endpoints relevant to tolerance, the program will not proceed to Phase 2. Pre-registered go/no-go criteria will govern that transition, and the open datasets generated by Phase 1 will remain a durable resource regardless of outcome. The likely scientific failure modes - effects too small to shift population risk, or heterogeneity too high to translate into general guidance - are each testable within Phase 1, before substantial clinical investment.

Theory of change. The activities in Phase 1 and Phase 2 are organized around an explicit theory of change: mechanistic discovery enables focused clinical trials, clinical evidence triggers pediatric society guidance, guideline movement and regulatory clarity unlock food industry investment, and accessible products reach families. The translational pathway followed by the LEAP trial, which generated clinical evidence and led to updated pediatric dietary guidelines within 2 years, is open to fermented foods. Each handoff is a hypothesis, and if any one breaks, the Initiative will learn where the real bottleneck lies. The fund's success will be measured not only by scientific output but by whether translation happens.

Why Philanthropy

Coordination across science, policy, and industry. The Initiative will oversee the full chain from foundational science to dietary practice, which no single institution - academic, industry, or government - is structured to do alone.

Prevention versus treatment economics. The pharmaceutical industry invests substantially in treatment for atopic disease, not prevention. The food industry has an active market for fermented and functional foods, but its business model funds product development and marketing, not the foundational research needed to establish prevention claims. The prevention case is therefore consistently underfunded by both, and presents a unique gap for philanthropy to fill.

Together, these features make philanthropy the right capital for this kind of problem, and the right capital to drive a durable, evidence-based shift in pediatric nutrition.

Leadership and Structure

Fund Lead: Dr. Rachel Dutton. I have spent the past 15 years working on the microbiomes of fermented foods, leading academic labs as a fellow at Harvard and as a Professor at UC San Diego. My team generated the first open multi-omic datasets linking fermented food composition to immune-relevant biology during my time leading the Microcosm Foods project at the Astera Institute. My network across academia, industry, and the nonprofit sector positions me to coordinate a thesis-driven fund of this shape. The Initiative will be guided by an interdisciplinary scientific advisory board.

The First Foods Initiative is now seeking philanthropic partners to anchor the fund.

Technical Appendix- selected supporting evidence

Childhood atopic disease prevalence and trends. Childhood food allergy prevalence increased ~50% between 1997 and 2011 and another ~50% between 2007 and 2021 (CDC/NCHS National Health Interview Survey; Branum & Lukacs 2008, NCHS Data Brief No. 10; Jackson et al. 2013, NCHS Data Brief No. 121). Pediatric eczema prevalence rose from 7.4% in 1997–1999 to 12.5% in 2009–2011 (Jackson et al. 2013). Peanut allergy incidence in one-year-olds tripled between 2001 and 2017 in U.S. health insurance claims data. As of 2024, 5.3% of U.S. children have a diagnosed food allergy and 12.7% have eczema (CDC/NCHS Data Brief No. 546, 2026, based on 2024 NHIS data).

Economic burden. Childhood food allergy in the U.S. is estimated at \$24.8B annually, including \$4.3B direct medical costs and \$20.5B family/opportunity costs (Gupta et al. 2013, JAMA Pediatrics).

Fermented foods and immune status. Wastyk et al. (2021), "Gut-microbiota-targeted diets modulate human immune status," *Cell*. A randomized trial in 36 healthy adults showed a high-fermented-food diet (~6 servings/day) increased gut microbial diversity and decreased 19 inflammatory markers across participants, effects not seen in the high-fiber arm.

Yogurt consumption and infant atopy (observational). Shoda et al. (2017), *Journal of Dermatological Science/ T-CHILD cohort*: habitual infant yogurt consumption associated with reduced atopic dermatitis (aOR 0.70) and food sensitization (aOR 0.53) at 5 years (n=1166). Tan et al. (2023), *Food & Function*: maternal yogurt consumption in late pregnancy associated with reduced infantile eczema (aRR 0.54 overall, 0.48 with >3x/week; n=2371). Crane et al. (2018), *Clinical & Experimental Allergy*: regular infant yogurt consumption associated with 50–80% reduction in eczema and atopic sensitization in a New Zealand cohort. Across multiple cohorts, effect sizes are consistent and dose-responsive.

Early oral introduction as a preventive paradigm. Du Toit et al. (2015), "Randomized trial of peanut consumption in infants at risk for peanut allergy," *NEJM*. The LEAP trial demonstrated an 81% relative reduction in peanut allergy at 5 years from early oral peanut introduction (4–11 months) versus avoidance. NIAID-sponsored guideline addendum followed in January 2017.

Tiered platform for early-life immune development. *Primary immune cell assays*: In vitro Th17 and Treg polarization from naive CD4+ T cells is a standard platform for screening dietary and microbial metabolites against immune cell fate decisions; recent examples include Hang et al. 2019, *Nature*, which used this approach to identify bile acid metabolites that bidirectionally control Th17 and Treg differentiation. Dendritic cell-T cell co-culture and transwell systems with cytokine readouts are similarly mature. *Infant-derived enteroid and immuno-organoid systems*: Adeniyi-Ipadeola et al. 2024, *mBio* 15:e01316-24, established that infant-derived intestinal enteroids have characteristic features of the immature infant epithelium; Sharafian et al. 2025, *Frontiers in Immunology*, demonstrated infant enteroid co-culture with neonatal Th17 cells to model epithelial-immune crosstalk during the developmental window. *Gnotobiotic and humanized-microbiota mouse models*: Arrieta et al. 2015, *Sci Transl Med*, used a humanized-microbiota mouse model to show that specific bacterial taxa missing from at-risk infant microbiomes reduce airway inflammation when restored. Two-generation HMA designs for studying allergy-relevant endpoints from infant donor microbiota have been published recently (de Jong et al, 2025, *BMC Microbiology*).

Gut microbiota, immune tolerance, and fermented weaning foods. On early-life microbiota and immunity: Belkaid & Hand (2014) *Cell*; Olin et al. (2018) *Cell*; Stewart et al. (2018) *Nature* (TEDDY). On global fermented weaning traditions (ogi/pap/koko, uji, togwa, mahewu): Ogunniran et al. (2024), *Comprehensive Reviews in Food Science and Food Safety*; FAO documentation of cereal-based traditional fermented foods:<https://www.fao.org/4/x2184e/x2184e07.htm>.

Datasets and prior work from the Microcosm Foods project, funded by the Astera Institute. The Initiative builds on a foundation of data and reagents developed to map the connections between fermented foods and immune biology (<https://zenodo.org/communities/microcosmfoods>), including multi-omic characterization of over 100 fermented foods and their impact on human immune cells (metabolomics data generated in collaboration with the Periodic Table of Food Initiative).